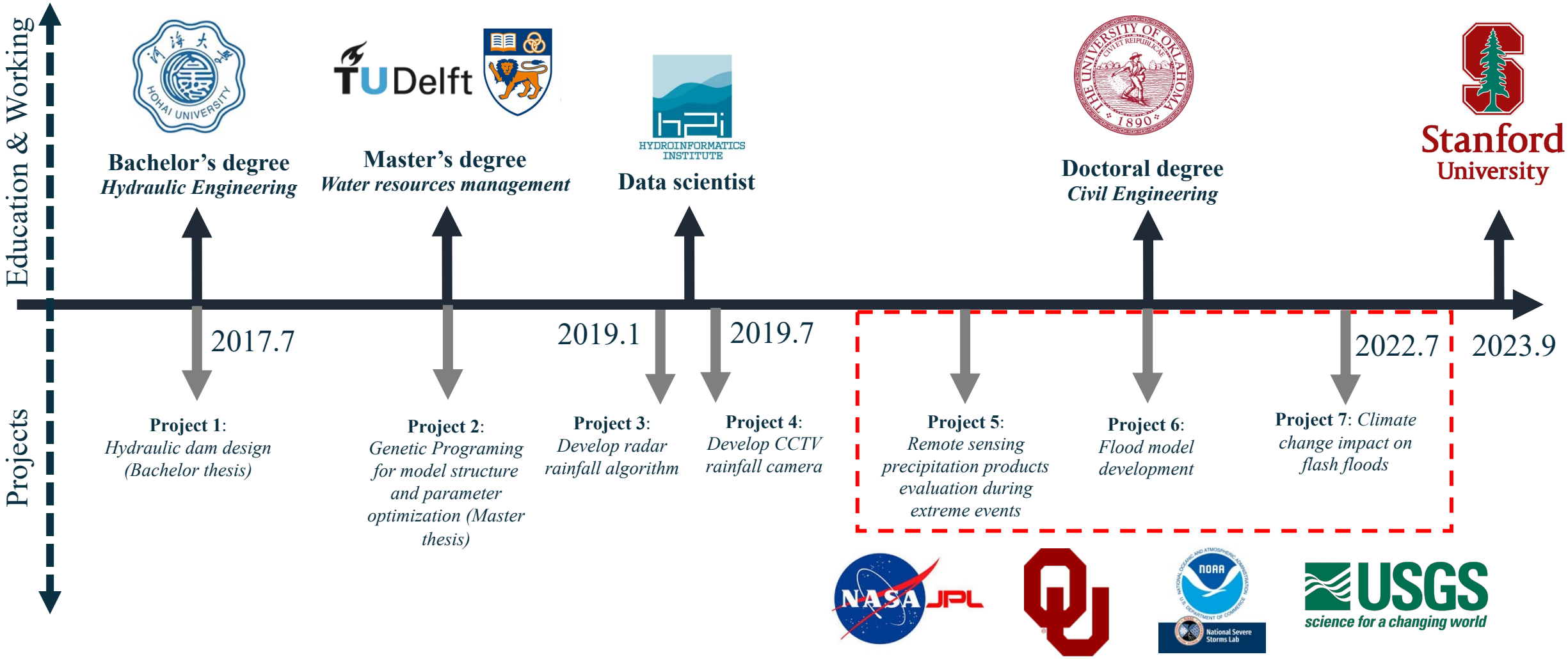
Advancing flood characterization through data, model, and application

Zhi Li, PhD

Stanford Doerr School of Sustainability Dean's Postdoc Fellow

Stanford University

Educational background and research experience

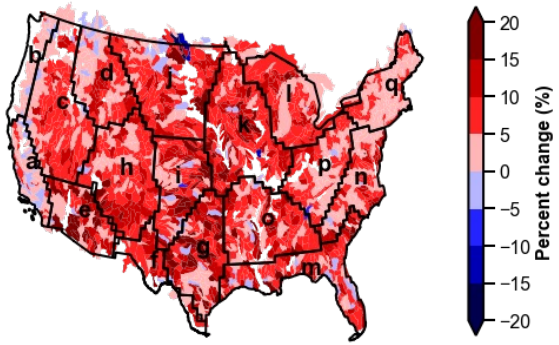


PhD dissertation: Advancing flood prediction and projection in a warming climate

1. **Remote sensing** monitoring water cycle (Li et al., 2021; Gao et al., 2021)
2. A **120-year flood database** to analyze flood risks (Li et al., 2021)



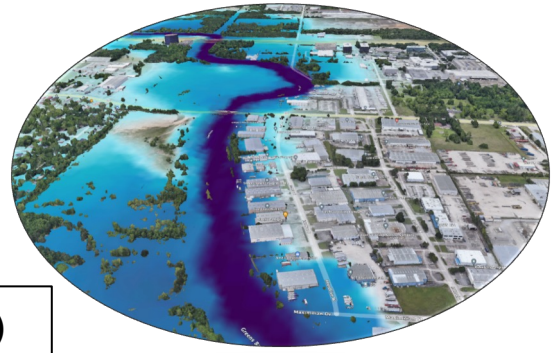




People-centered



1. Developed a **hydrologic-hydraulic coupled model** for better flood prediction (Li et al., 2021,2022)
2. Developed a vector-based hydrologic model for **large-scale** streamflow prediction (fast and high resolution) (Li et al., 2022)

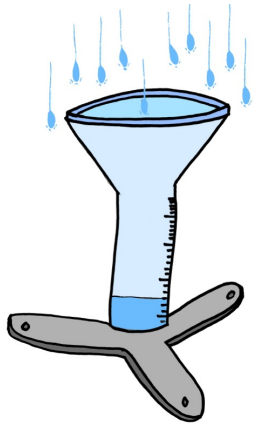


1. Investigated **future flash floods** (Li et al., 2022)
2. **Spatiotemporal characteristics** of future floods (Li et al., 2022)

1.1 Quantitative Precipitation Estimates by instruments are uncertain

Why we care about precipitation data errors?

Errors in precipitation products can propagate to streamflow prediction, resulting in even larger errors in streamflow.

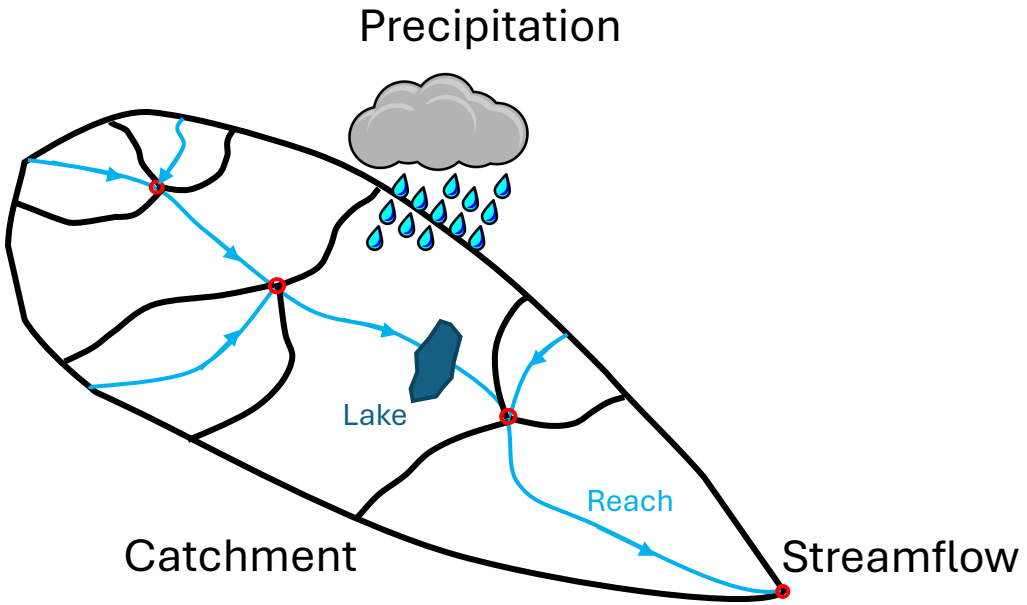




Z-R relation and difficulty identifying hydrometeors



Indirect and infrequent estimates



1.2. We need a unified multi-source flood database

Issues in existing flood database:

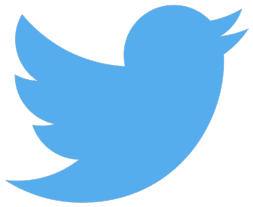
1. Single sourced and centralized
2. Non-uniformly distributed
3. Not up-to-date

We propose a decentralized, multi-source, and up-to-date flood database for model validation and flood risk studies

Flood reports



Social media (crowdsourcing)



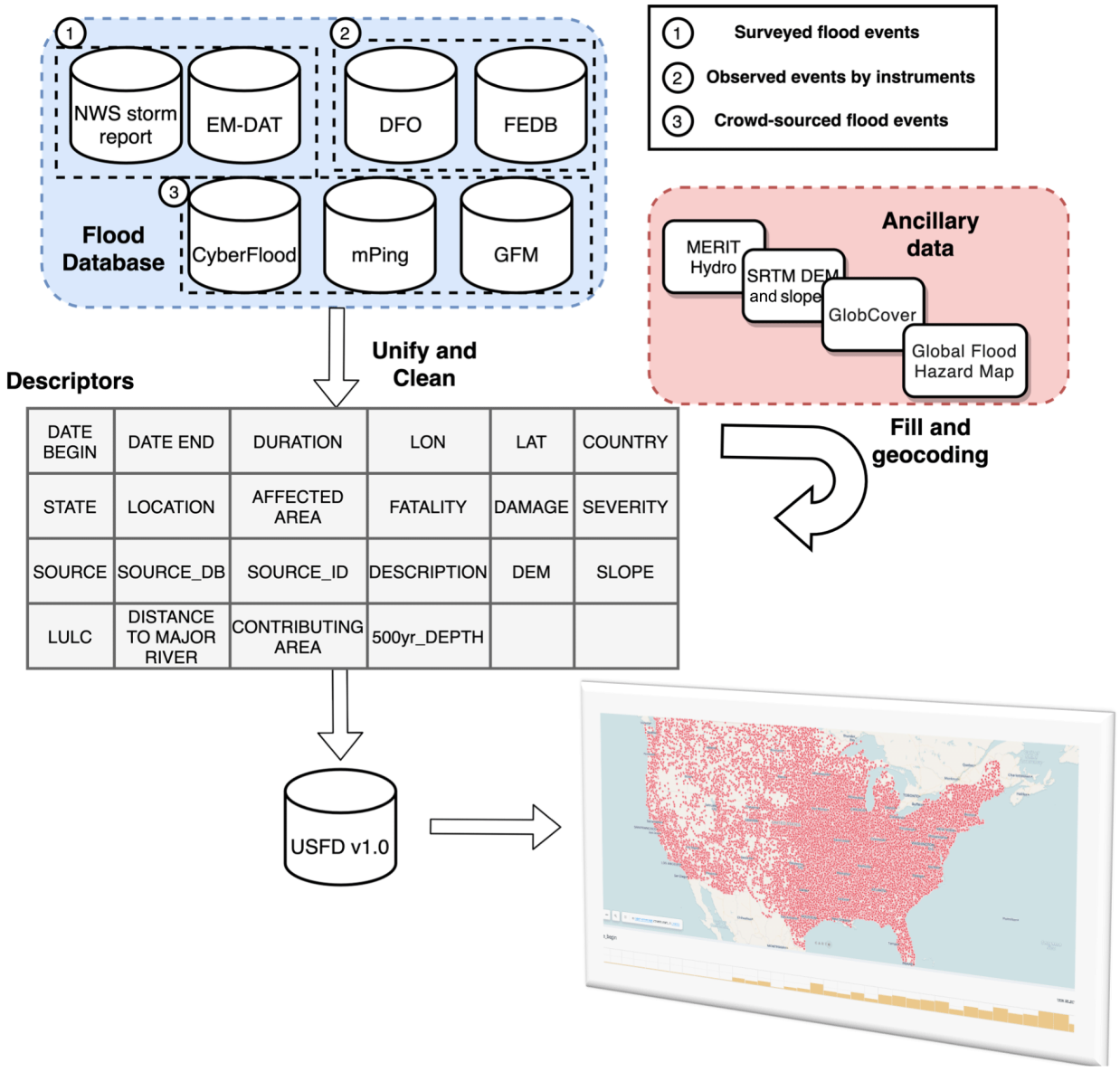
Satellite observation



Stream gauge



Processing methods – harmonization, unification, cleaning, and filling



In total, we recorded 698,507 flood events over the continental US from 1900-2020

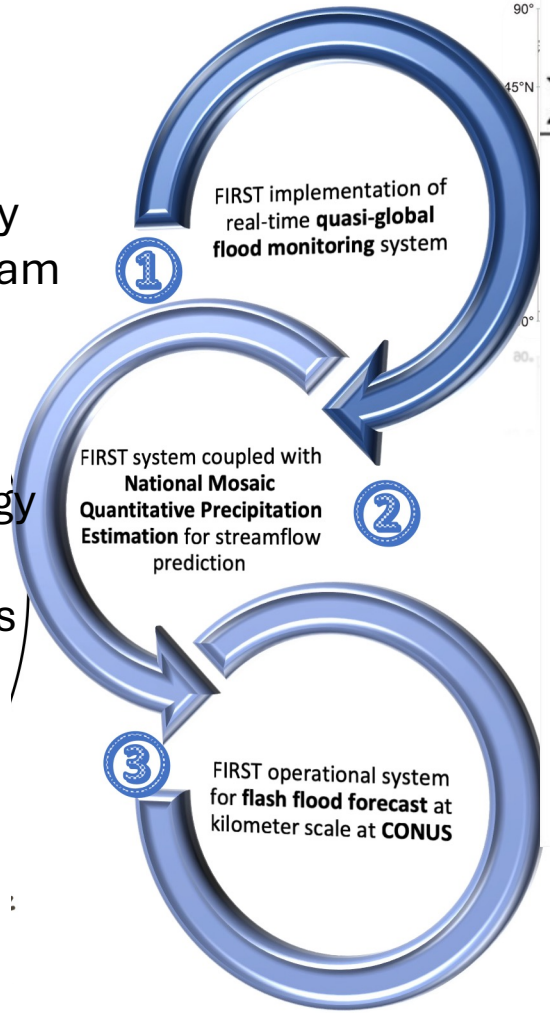
Li et al., 2021, ESSD

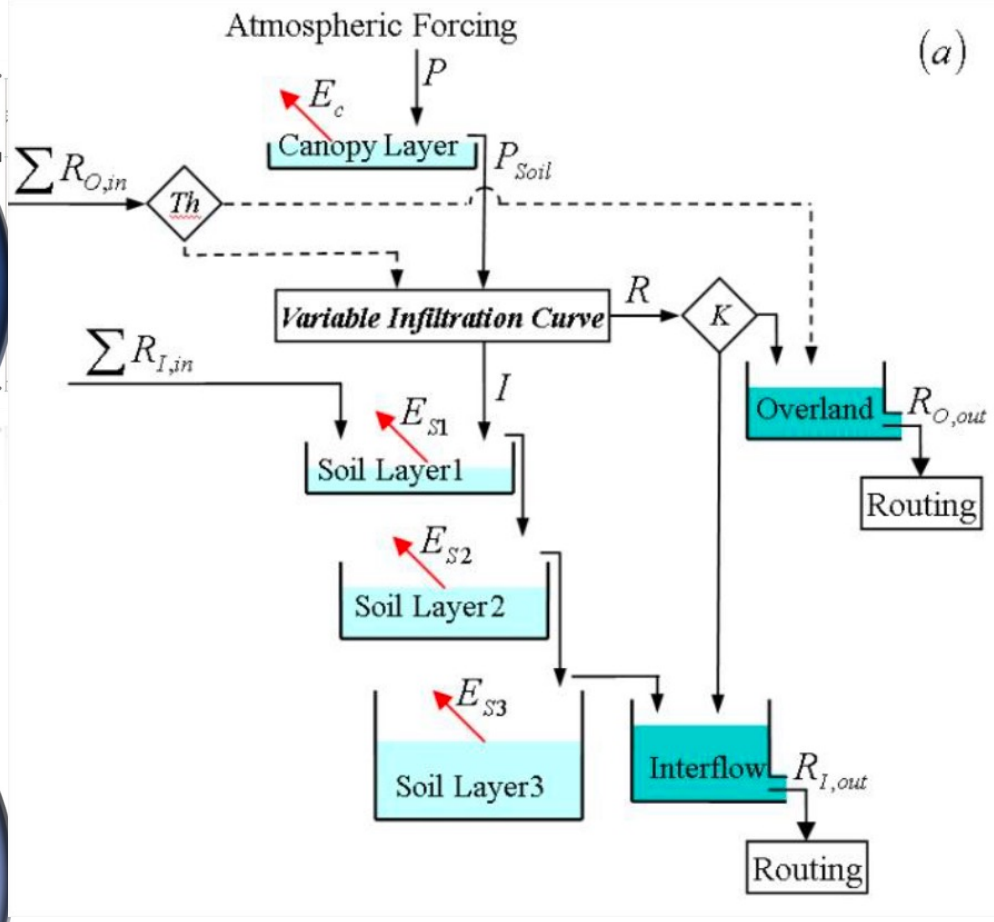
2.1. Decadal review of our in-house hydrologic model

The **C**oupled **R**outing and **E**xcess **ST**orage (**CREST**) distributed hydrological model, launched in 2011, is a hybrid modeling strategy developed by OU and NASA SERVIR Project Team (Wang et al., 2011).

Features:

- Spatiotemporal variation of water and energy flux and states
- Global hydrological prediction at fine scales
- integration of remote sensing data sets
- Efficient



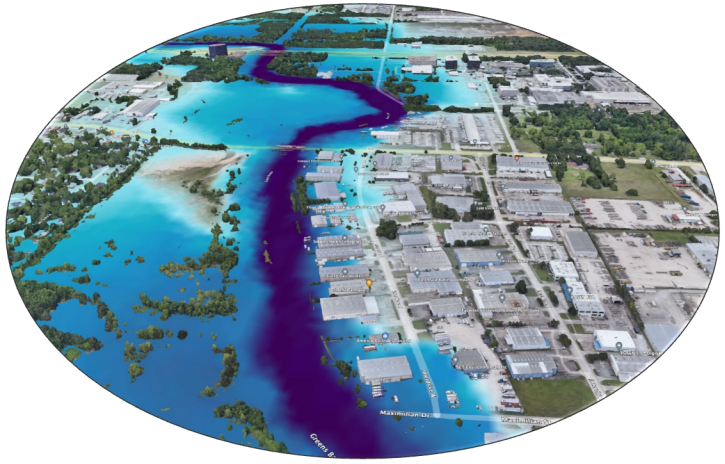


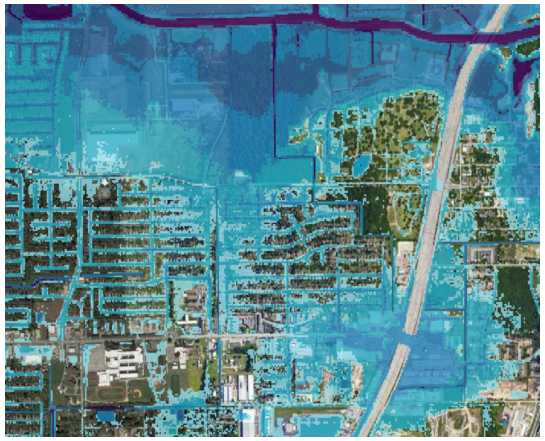
CREST V1.X (Wang et al., 2011)

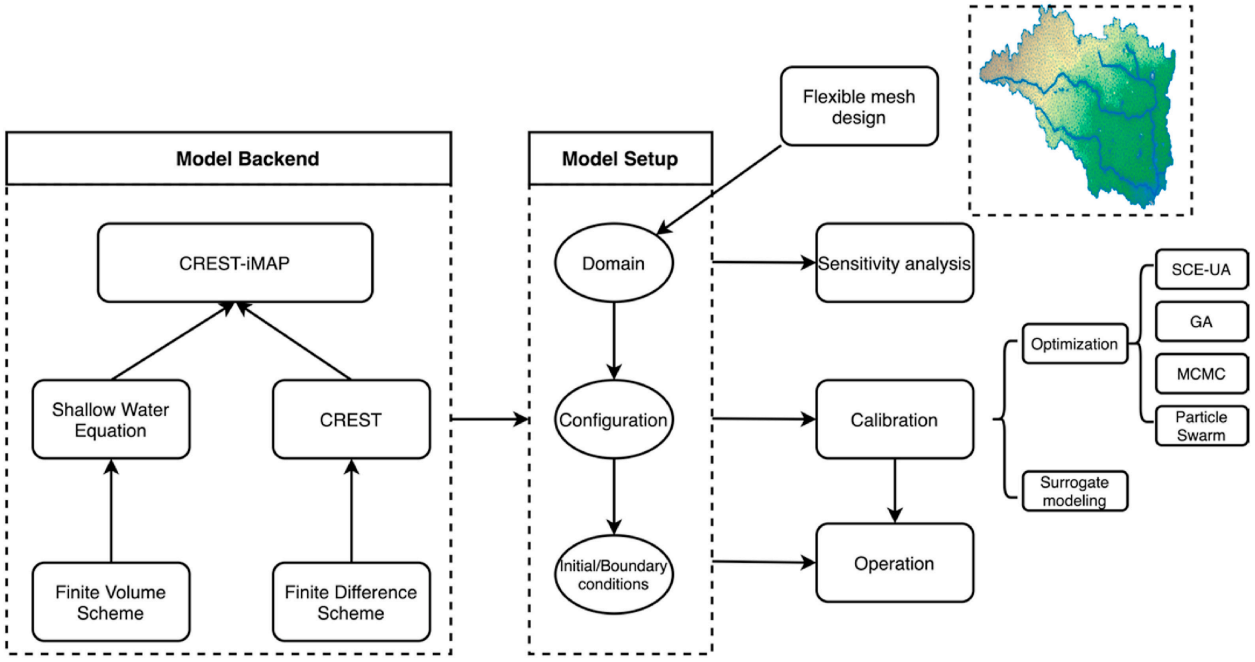
Li et al., 2022. J. Hydrol

2.2. A newly designed hydrologic-hydraulic framework

CREST-iMAP stands for Coupled Routing and Excess Storage inundation MApping and Prediction. It couples CREST model with a 2D shallow water solver to predict **flood depth** and **velocity**.







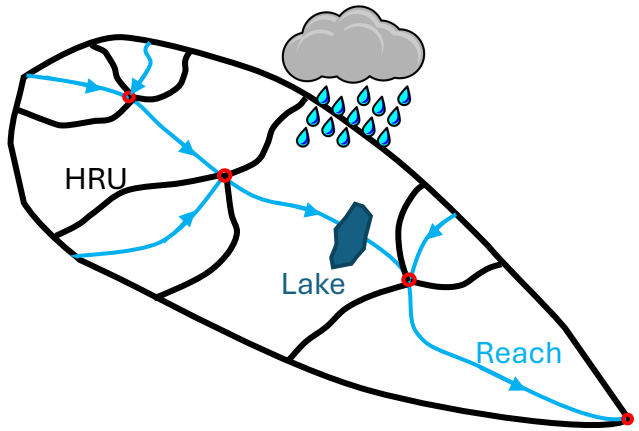
https://github.com/chrimerss/CREST-iMAP

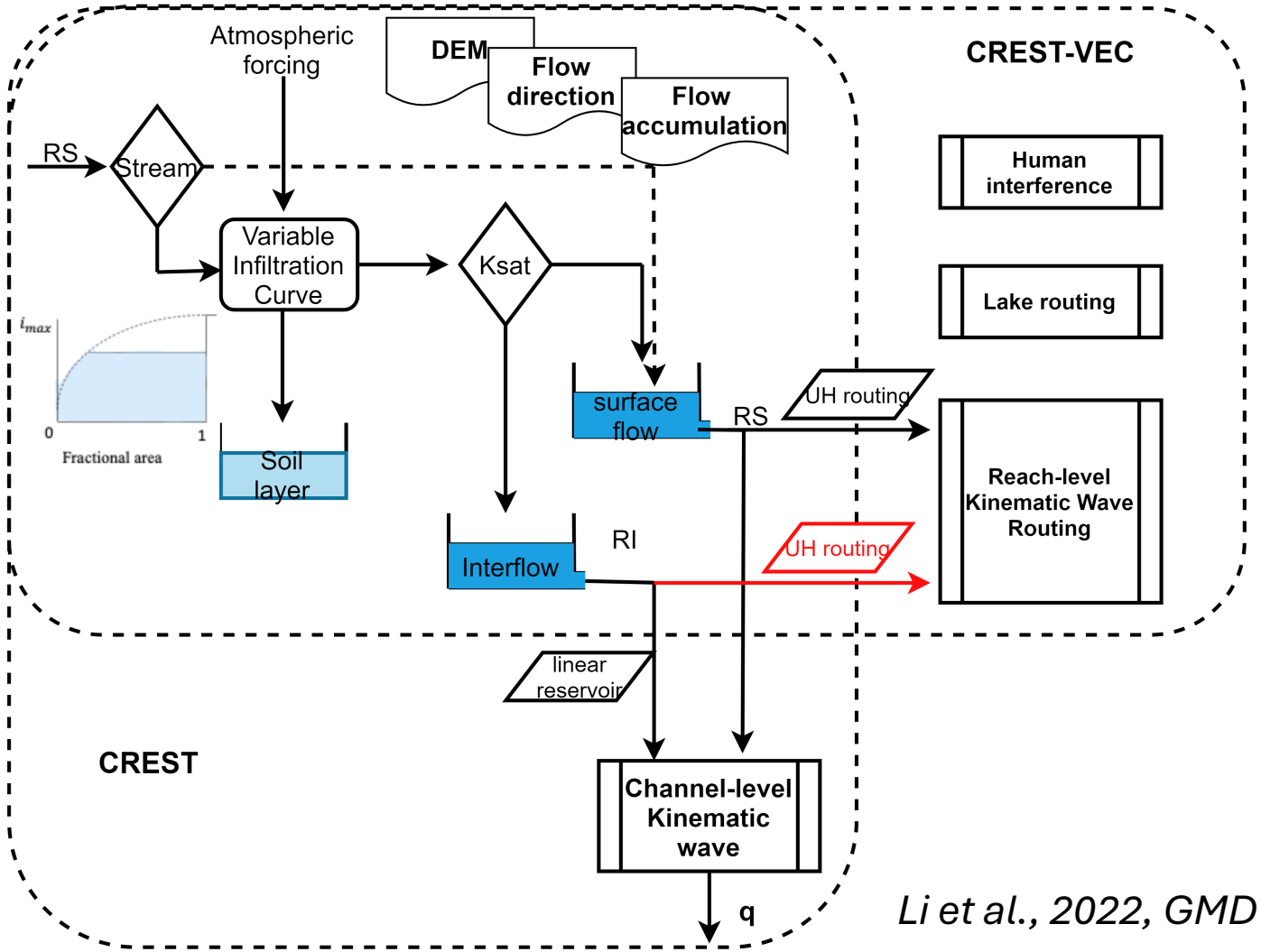
Li et al., 2021, EMS

2.3 A vectorized modeling framework – CREST-VEC

Features:

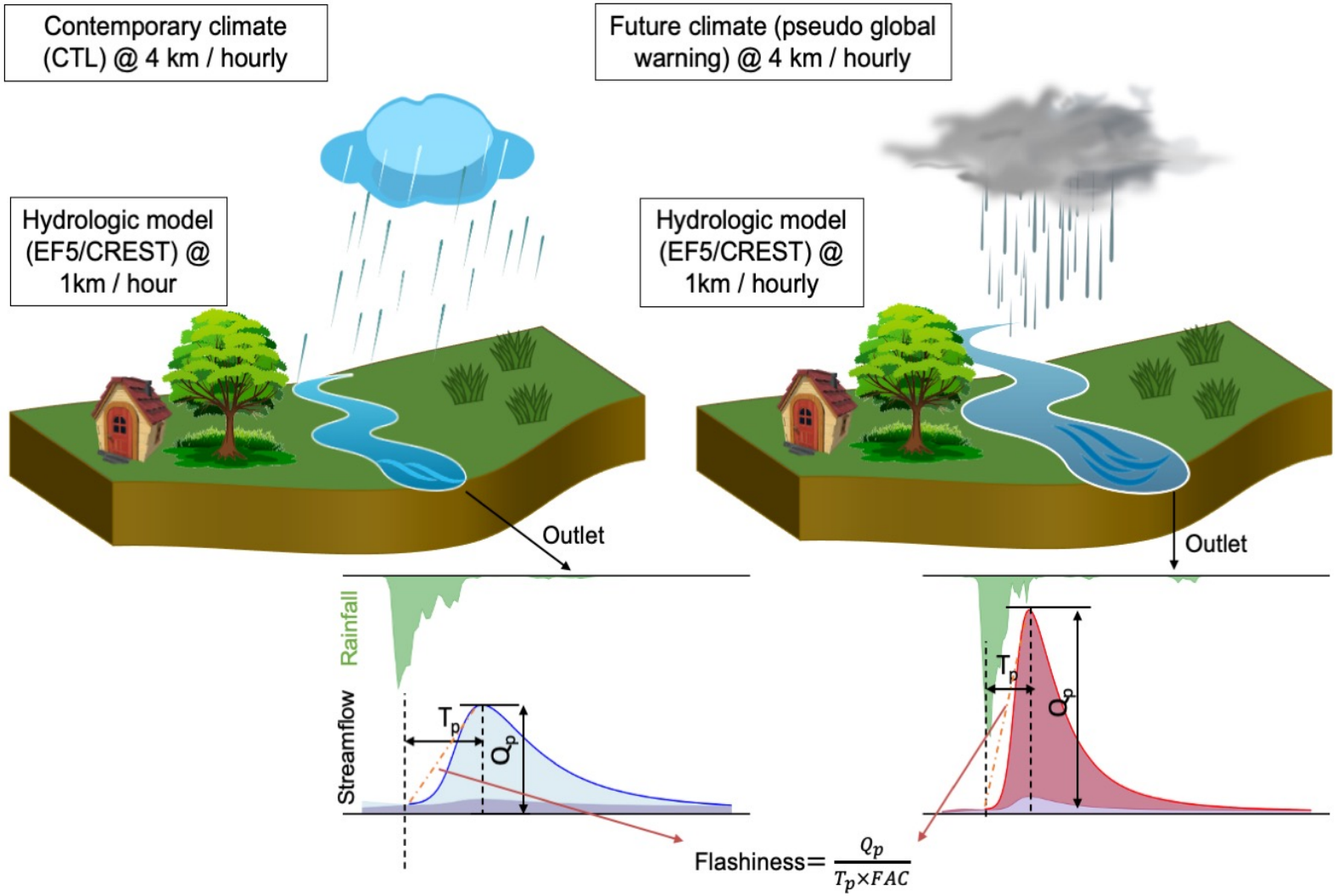
1. Routing basin surface runoff and subsurface runoff at hydrologic response unit based on unit hydrograph
2. Accumulate streamflow at each river reach and route based on unit hydrograph or kinematic wave
3. Lake simulation to regulate river flow



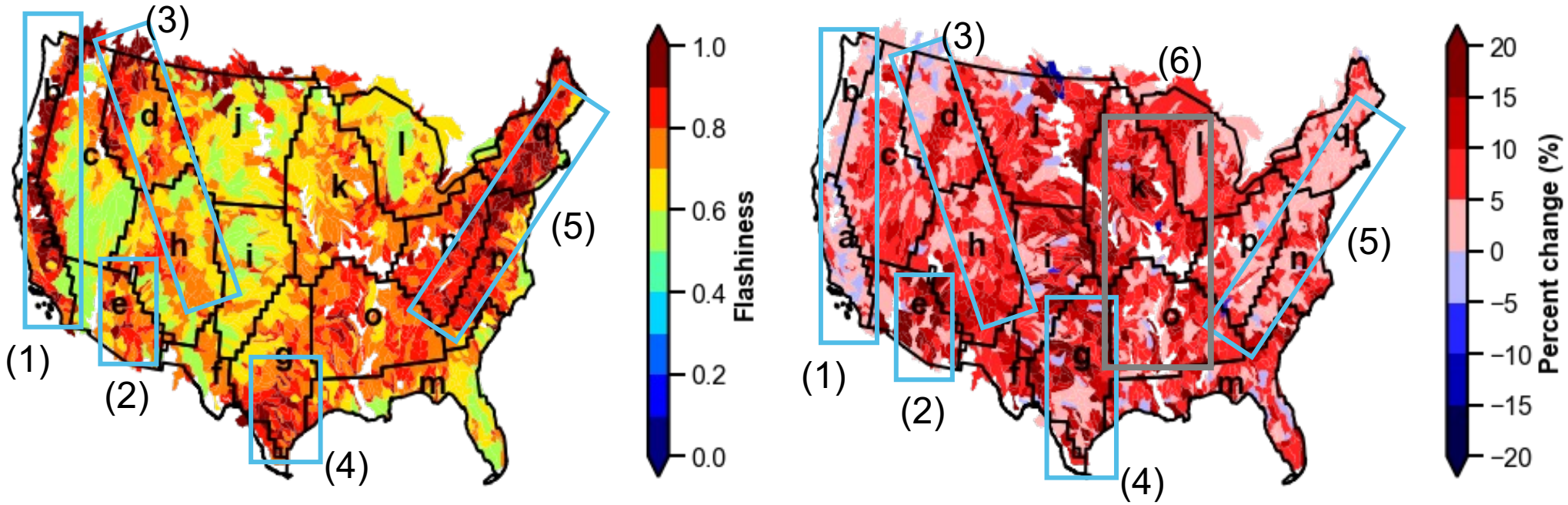


Li et al., 2022, GMD

Flood risk in a changing climate



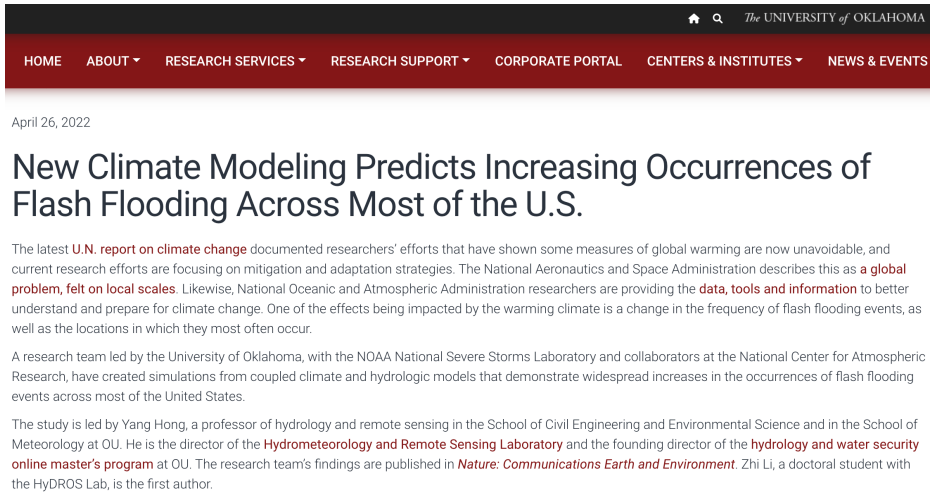
US floods are becoming flashier in the future



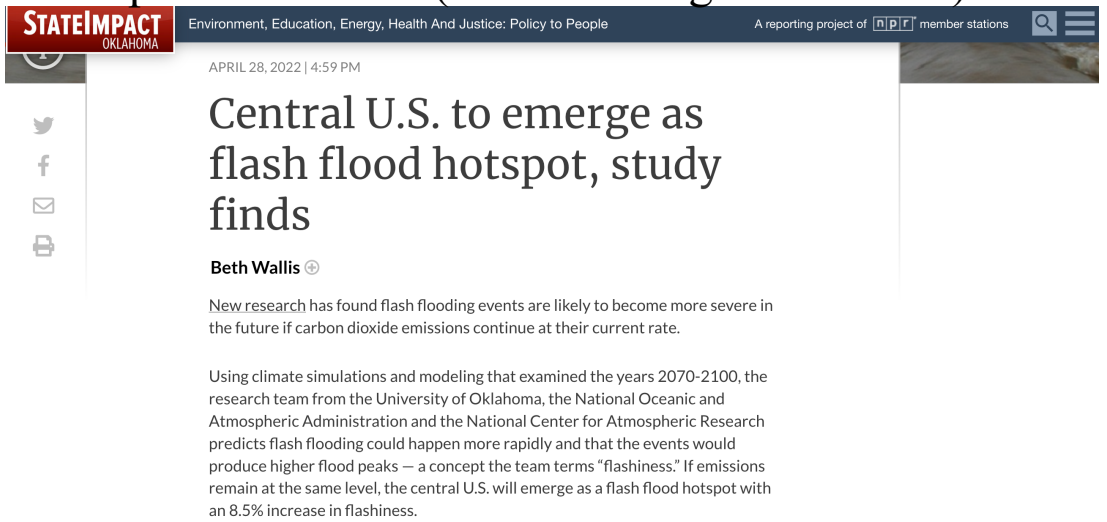
1. US floods are becoming **7.9%** flashier by the end of the century assuming a high-emissions scenario
2. The Southwest (+10.5%) has the greatest increase in flashiness among historical flash flood hot spots, and the central US (+8.6%) is emerging as a new flash flood hot spot.
3. future flash flood-prone frontiers are advancing northwards .

Our flood work has been featured by local, national, international news media

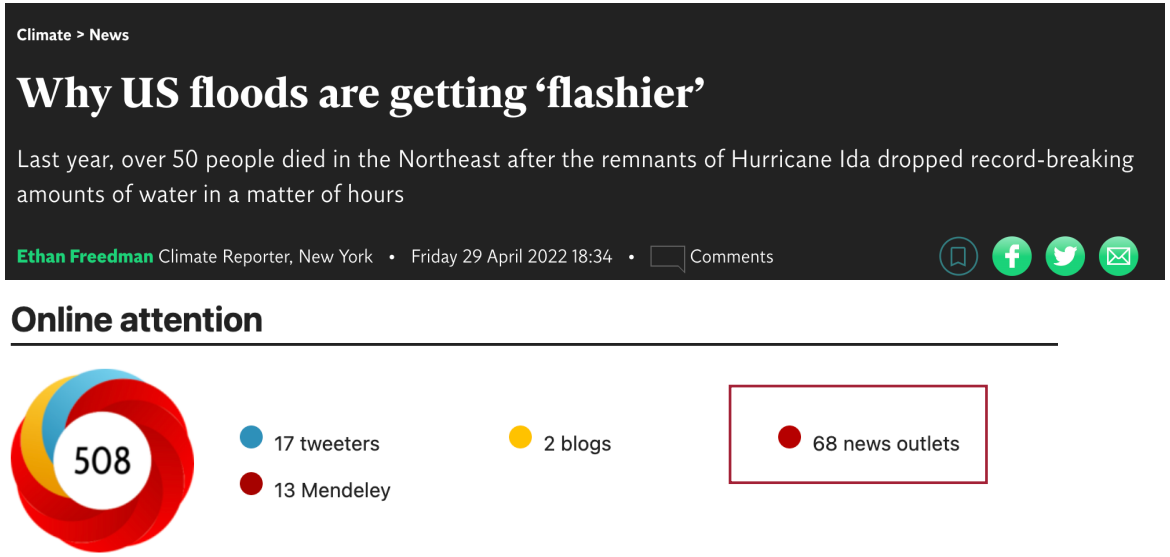
OUVPRP



StateImpact Oklahoma (NPR all things considered)



Independent (UK based)



The New York Times



Current work

- Flood impact on food security
- Flood impact on public health
- Deep Learning weather foundation models

Service

- Associate Editor – Journal of Applied Meteorology and Climatology
- Invited proposal reviewer by NASA
- AGU primary convener
- Invited reviewer for over ten journals